Structures & states

Describe particles

1 Solid → liquid → gas

Solid: fixed arrangement, vibration. Liquid: close particles move past one another. Gas: widely spaced, rapid random motion.

Melting/freezing; boiling/condensing. Heating transfers energy; changes of state are physical.

2 Read melting & boiling data

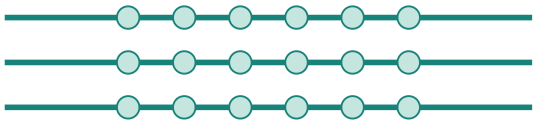
Below melting point → solid. Between melting and boiling → liquid. Above boiling → gas.

State symbols: (s), (l), (g), (aq). Aqueous means dissolved in water.

Particles are not hard identical spheres: models omit forces, size differences and continuous movement.

Explain properties

3 Metals conduct & bend



Layers can slide past each other

Strong attraction between positive ions and delocalised electrons → giant metallic structure.

Electrons move and carry charge and transfer energy. Metal layers slide while metallic bonding remains.

4 Alloys resist bending

An alloy mixes a metal with other elements. Different-sized atoms distort regular layers.

Layers slide less easily → alloy is harder. Use a labelled particle drawing to explain, not just "stronger".

Compare models

5 Polymer chains

Many atoms joined by strong covalent bonds in long molecules. Forces act between chains.

Intermolecular forces are usually sufficient for polymers to be solid at room temperature.

A repeating unit in brackets with n represents many linked units.

6 Recognise giant structures

Diamond, graphite and silicon dioxide are giant covalent. Many strong bonds need much energy to break → high melting.

Contrast small molecules: melting/boiling does not break their internal covalent bonds.

Use diagrams, formulae and melting/conductivity data together to identify structure.